Is more thinking always better? Think again: Model-cognition fit in AI-supported sensory analysis

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Abstract

Large language models (LLMs) can score open-ended consumer text, and many current systems expose a configurable *thinking* setting: an inference option that spends extra compute on intermediate reasoning before the model returns a final answer. More deliberative settings are often assumed to score better. We tested that assumption on the published Ideal-Free-Comment study in which 483 French consumers evaluated 30 commercial cooked hams at home (2,758 evaluations), describing each product and their ideal ham in their own words while also rating liking. Using typical fast commercial models from one family available at the time of the experiment, we compared each consumer's actual and ideal Free-Comment descriptions and returned structured alignment scores for visual appearance, texture, and flavor, which were then used to predict liking under consumer-group holdout. Across 27 configurations that varied direct-response versus thinking level, response scale size, and temperature, a direct-response setup predicted liking best ($R^2 = 0.573$), clearly above a generic embedding-similarity baseline. Setups with thinking turned on, from a related generation in the same family, were slower and did not close that gap. In a controlled comparison at six scale points and temperature 0.7, they were also less reliable at returning parseable structured output. Within that thinking-capable generation, low thinking improved prediction and structured-output reliability over minimal thinking but substantially increased runtime. The alignment scores recovered the published sensory modality hierarchy and, in a held-out construct check, recovered Just-About-Right ratings the model never saw, attribute by attribute; thinking settings did not improve that recovery. These results come from one model family (Gemini fast commercial models available when we ran the experiment). They are not general laws about all LLMs. The point of the paper is that thinking level matters for sensory text scoring, and that it can matter in a surprising way: more deliberation was not automatically better. A careful analogy to ideas about fast versus more deliberate human judgment offers one way to think about why that pattern might arise. The human brain is far more complex than any current LLM, but the parallel may help frame new work in sensometrics. Readers should treat our findings as a prompt to try thinking settings on their own tasks, whether they use closed or open models.

Keywords: Free-Comment; Ideal-Free-Comment; sensory analysis; consumer liking; large language models; model-cognition fit.

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1 Introduction

Consumer sensory research needs to measure how products compare with what consumers actually want, and it needs to do so at scale. Structured elicitation methods such as Check-All-That-Apply (CATA), Just-About-Right (JAR) scales, and the Ideal Profile Method (IPM) give analysts clean tables, but they impose a vocabulary in advance [Worch et al., 2013]. Free-Comment methods do the opposite. They let consumers use their own words, which brings the data closer to the consumer's own experience while avoiding vocabulary restriction, awareness effects, dumping effects, and acquiescence bias. The cost is a text-processing problem: free comments must be cleaned, normalized, grouped, and transformed into structured variables before they can be modeled.

Free-Comment has a clear place in sensory science. ten Kleij and Musters [2003] introduced open-ended response analysis as a complement to preference mapping. Symoneaux et al. [2012] showed how comments about likes and dislikes could support preference analysis. Mahieu et al. [2020] found that Free-Comment outperformed CATA for consumer wine characterization in a home-use setting, and Mahieu et al. [2021] developed a multiple-response chi-square framework and Multiple-Response Correspondence Analysis (MR-CA) for these data.

Ideal-Free-Comment (IFC) extends the idea by adding the consumer's own ideal. In the consumer study of cooked hams by Mahieu et al. [2022], consumers described the actual hams they evaluated, rated their liking of each ham in the same sessions, and then described their ideal ham directly, using the same three sensory modalities: visual appearance, texture in mouth, and flavor. The method is related to IPM [Worch et al., 2013], which asks consumers to rate perceived and ideal intensities on a fixed attribute list, and to Free-JAR profiling [Luc et al., 2020], which keeps the Just-About-Right logic but collects it through open descriptions. IFC keeps the actual-versus-ideal logic while replacing the fixed attribute list with consumers' own words. Mahieu et al. [2022] showed that drivers of liking and ideal-product descriptions provide complementary information, and that ideal descriptions can reveal consumer segments. This paired design creates an opportunity: if each consumer gives both actual and ideal comments, the central comparison shifts from whether a descriptor appears to whether the actual product matches that consumer's ideal experience.

Processing such comments has so far required manual expertise. Text mining has been used in sensory science for lexicon development, descriptor extraction, term grouping, and analysis of reviews or open comments [Hamilton and Lahne, 2020, Miller et al., 2021, Hamilton et al., 2026], and earlier work treated automated sentiment of Free-Comment as an indirect liking measurement [Visalli et al., 2020]. Recent work treats preprocessing itself as a methodological variable [Visalli and Mahieu, 2023, Visalli et al., 2025b] and asks directly whether natural language processing or large language models (LLMs) can replace human operators for that preprocessing [Visalli et al., 2025a]. Machine learning has also begun to complement classical preference mapping for identifying drivers of liking [Rios de Souza et al., 2024].

LLMs are now widely used for text evaluation, annotation, and semantic scoring, and a parallel line of work uses them as simulated respondents in social, economic, and market research [Argyle et al., 2023, Horton, 2023, Brand et al., 2023]. Work in the literature often labeled LLM-as-a-judge shows that LLM evaluation and annotation can be useful but not automatically reliable [Zheng et al., 2023, Liu et al., 2023, Gilardi et al., 2023]. Known issues include positional bias, verbosity bias, self-enhancement bias, calibration problems, and task-specific disagreement with human raters. In sensory and consumer research, recent work has considered generative AI across concept, design, and testing phases [Motoki et al., 2025], coffee flavor annotation [Jakkaew et al., 2026], flavor-pairing creativity [Wang and Pellegrino, 2025], exploratory use of ChatGPT as a sensory evaluator [Torrico, 2025], and whether LLMs reproduce human food choices in discrete-choice experiments [Califano and Caracciolo, 2026].

This paper studies a method we call *direct LLM alignment scoring*: the model receives a consumer's paired actual and ideal Free-Comment descriptions and returns an integer alignment score for each sensory modality, indicating how closely the actual experience matched that consumer's ideal. The resulting features are small, interpretable, and ready for tabular prediction of liking. The scope is narrower than general LLM judging. The model does not rank open-ended product concepts or evaluate arbitrary prose. It receives a fixed comparison task, fixed sensory modalities, a fixed scale, and a strict response schema in JavaScript Object Notation (JSON). That narrower setup makes the method easier to audit, but it does not remove the need for validation. Semantic Similarity Rating (SSR) is an important adjacent method, but it is not the method used here: in SSR the LLM produces free-text responses that are mapped to Likert probability distributions through embeddings of reference anchor statements [Maier et al., 2025]. Direct alignment scoring instead asks the model for the integer scores themselves. It is simpler, cheaper, and easy to parse, but it does not provide the distributional information SSR can provide, and direct numeric scoring can compress response variation if the model overuses the middle of the scale.

Terminology. In this paper, *thinking* follows current LLM product usage: a controllable inference setting in which the model spends extra compute on intermediate reasoning before returning a final answer [OpenAI, 2024; Google, 2026; Wei et al., 2022]. It is not a claim that the model thinks like a person, and it is not a permanent model type. The same base model can be run at different thinking levels, and providers expose the setting as an explicit option (for example `thinkingLevel` in the Gemini API). We use that product term because it is the setting researchers actually choose. We call setups that answer without this extra deliberation *direct-response* (or non-thinking) configurations.

Whether extra thinking helps depends on the task. Chain-of-thought prompting and reasoning-oriented configurations improve many tasks that require explicit decomposition [Wei et al., 2022]. More recent work shows that extra reasoning can also hurt. Gema et al. [2025] report inverse scaling with test-time compute in settings where irrelevant content, spurious features, or loss of focus can mislead a model. Zhou et al. [2026] analyze cases where extended reasoning changes an initially correct answer into an incorrect one. Liu et al. [2024] report that chain-of-thought can reduce performance on pattern-matching or perceptual tasks. In a meta-analysis of more than 100 studies, Sprague et al. [2025] find that chain-of-thought helps mainly on mathematical and symbolic reasoning and adds little on other task types. Sui et al. [2025] review overthinking and efficient reasoning more broadly. A related line of work finds that the reasoning a model verbalizes can be unfaithful to the computation that produced its answer [Turpin et al., 2023]. That is one reason added reasoning steps need not improve a constrained scoring task.

Sensory judgments are a natural place to test this task dependence. Consumers do not usually decide whether they like a food by reasoning through a long checklist. They see it, taste it, feel its texture, and react. The underlying judgment is fast, sensory, and affective. As an analogy, not as a claim about model internals or neural implementation, this contrast recalls dual-process ideas about fast, automatic judgment and slower, more deliberative reasoning [Kahneman, 2011, Evans and Stanovich, 2013]. Direct-response configurations return answers without extended intermediate reasoning; thinking configurations spend more test-time compute on that intermediate stage. If the human target is closer to a fast affective judgment than to deliberate analysis, a direct-response setting may be the better match for the scoring job. We call this idea *model-cognition fit*: choosing the setup whose style of response, fast versus deliberative, is better matched to the judgment being measured. The human brain is far more complex than any current LLM. The analogy is still useful: it suggests why thinking level might change measurement quality in a surprising way, and it points sensometricians toward new research on how well a scoring setup fits the task, rather than on model brand names alone. The practical question is whether researchers should treat thinking as a default upgrade or as a setting to try out first.

We tested that idea on the cooked-ham IFC dataset of Mahieu et al. [2022] (described in Section 2.1), scoring all 2,758 consumer-product evaluations under a range of LLM setups that varied thinking style, response scale size, and sampling temperature within one family of fast commercial models available when we ran the experiment. The paper asks five research questions:

1. How do direct-response and thinking-enabled LLM configurations compare on sensory actual-versus-ideal alignment scoring, in predictive accuracy, speed, and structured-output reliability?
2. Can direct LLM alignment scores predict consumer liking better than a generic embedding-similarity baseline?
3. Do the resulting features recover the sensory hierarchy reported by Mahieu et al. [2022]?
4. Do the alignment scores recover independent structured judgments, the held-out JAR ratings, that the model never saw?
5. How should direct alignment scoring be positioned relative to Free-Comment analysis, IFC, and SSR?

The aim is not to rank commercial models. Named application programming interface (API) models change faster than peer review, and any ranking is a snapshot. The aim is to test whether model settings matter for sensory scoring and to draw the methodological consequence: pilot task-matched configurations before locking a pipeline.

2 Materials and methods

2.1 Dataset and reference analysis

The case study uses the cooked-ham data of Mahieu et al. [2022], with the open dataset documented by Visalli et al. [2024]. The study included 483 French consumers from seven cities. Consumers purchased and evaluated cooked hams at home over 13 weeks. The product set contained 30 commercial cooked hams selected to cover variation in fat and salt content in the French market. The set excluded smoked, braised, spit-roasted, flavored, and rind-on products.

Each consumer evaluated one or more hams; in the original paper, consumers evaluated between 1 and 14 hams, with a mean of 5.71. The source workbook used here contains 2,758 product evaluations. Throughout this paper, the unit of analysis is one such *evaluation*: one consumer's assessment of one product. For each evaluation, the consumer described the product by visual appearance, texture in mouth, and flavor in free text, rated overall liking on a 0–10 visual analogue scale, and gave structured JAR ratings for color, fat, salt, and tenderness, coded from -2 (really not enough) through 0 (just about right) to $+2$ (really too much). At the end of participation, the consumer described their ideal ham using the same three modalities. The JAR ratings were never shown to the model. We hold them out as a prompt-held-out structured validation signal from the same study and use them only for the internal construct check described in Section 2.6. Table 1 summarizes the dataset.

Table 1: Cooked-ham dataset used in the present analysis.

Quantity	Value
Consumers	483 French consumers
Products	30 commercial cooked hams
Evaluations	2,758 home-use evaluations
Product period	13 weeks
Liking measure	0–10 visual analogue scale
JAR ratings	Color, fat, salt, tenderness; -2 to $+2$ (held out for validation)
Text fields	Actual comments per evaluation; ideal comments per consumer
Original source	Mahieu et al. [2022]; open data article by Visalli et al. [2024]

The original analysis is the reference point for our validation. Mahieu et al. [2022] processed actual-product Free-Comment data separately by sensory modality. The French text was cleaned, lemmatized, filtered, and grouped into latent descriptors. Descriptors were retained when they were cited in at least 5% of evaluations for at least one product, and the retained descriptors were cross-tabulated by consumer and product, giving a descriptor-presence matrix. Liking was then modeled as a function of product, descriptor presence, and consumer, with product and descriptor effects as fixed effects and consumer as a random effect. Positive flavor descriptors such as fragrant, smoked, and ham taste increased liking. Negative descriptors such as insipid, elastic or rubbery, dry or pasty, and visible fat decreased liking. The broad hierarchy was clear: flavor had the strongest relation to liking, texture followed, and visual appearance was weaker. Ideal-product descriptions were analyzed by descriptor citation proportions, and Mahieu et al. [2022] also applied MR-CA by modality to product-by-descriptor citation proportions. We do not try to reproduce every descriptor effect. We ask whether direct LLM alignment scoring recovers the main sensory structure and predicts liking.

2.2 LLM scoring design

2.2.1 Models, thinking settings, and sampling parameters

The scoring design crossed three model configurations, three response scale sizes, and three sampling temperatures, for 27 direct-scoring configurations (Table 2). We stayed within one family of commercial fast models (Google Gemini configurations available at scoring time; notebook and grid run dated 15 January 2026) so that we could vary thinking style while holding the rest of the setup fixed. That is a scope choice for exploring thinking as a setting, not a survey of the market. The three configurations are examples from that family, not an endorsement of named products. In plain terms they are:

1. a *direct-response* configuration with no thinking setting;
2. the same thinking-capable generation at `thinkingLevel = "minimal"`;
3. that generation at `thinkingLevel = "low"`.

The last two share one base model, so they form a within-model thinking gradient. Higher thinking levels were not tested. For auditability the archive stores the API aliases used at call time (`gemini-flash-lite-latest` for the direct-response setup; `gemini-3-flash-preview` for the two thinking setups). The provider did not return a separate immutable model revision identifier in the saved response metadata. What matters for audit is the setting used (direct-response versus thinking level), temperature, scoring date, and the per-evaluation scores, not the brand label. Named systems change quickly; the question for science is whether the thinking setting changed measurement quality on this task.

This design tests the paper's central hypothesis. If sensory alignment scoring behaves like a fast judgment task, direct-response configurations should score it well. If extra deliberation improves the measurement, the thinking configurations should outperform the direct-response configuration.

Temperature controls the randomness of token sampling during generation: at temperature 0 the model almost always picks its highest-probability token, while higher values spread probability over more tokens and produce more variable output. We tested 0, 0.3, and 0.7 to span deterministic scoring, a mildly random setting, and a commonly used default region, because scoring workflows in practice differ in this setting. The nucleus and top-k sampling parameters (`topP`, `topK`) and the candidate count were left at the provider defaults, with one candidate response per call. Generation requested JSON output (`responseMimeType = application/json`) and used the assigned temperature and, where applicable, the assigned thinking configuration. One scoring job was initiated per evaluation per configuration, with up to three API attempts when generation, transport, or schema validation failed (including exponential backoff on HTTP 429).

Table 2: Experimental factors in the LLM scoring grid.

Factor	Levels varied	Held fixed
Model configuration	Direct-response; minimal thinking; low thinking (one family of fast models at the time of scoring)	Prompt task and JSON schema
Scale size	4-point, 6-point, 8-point scales	Actual-versus-ideal comparison
Temperature	0.0, 0.3, 0.7	topP, topK, candidate count (provider defaults)
Validation	Single grouped consumer holdout; 200-resample bootstrap for six selected configurations	No consumer overlap between train and test

2.2.2 Response scales

Each configuration asked for integer alignment scores on an ordered similarity scale with 4, 6, or 8 points; for example, the six-point scale ran from 1 = "Extremely Different" to 6 = "Extremely Similar" (the full label sets are given in A). These are similarity scales relative to the consumer's ideal, not liking scales. All three scales have an even number of points, so no single center category was offered and each modality score had to fall on the similar or the different side of the consumer's ideal. That choice is especially relevant for LLM scoring: LLM raters show human-like central-tendency bias on rating scales, clustering responses near the middle [Rupprecht et al., 2025], and forcing a side converts that compression into a directional decision. Because the scores enter a liking prediction model, a noncommittal center category would also add little discriminating information. Omitting the midpoint is likewise an established forced-choice device in human survey research [Garland, 1991]. We note that the bipolar labels are imperfect: an even scale still has near-middle categories such as "Slightly Similar." The parser's midpoint default, described in Section 2.2.4, applies only to evaluations that failed to return a usable score. It was not a response option offered to the model.

2.2.3 Scoring prompt and response format

Each evaluation was presented to the model as a paired actual-versus-ideal comparison. The prompt contained six evaluation-specific text fields: the consumer's ideal visual, texture, and flavor comments and the actual visual, texture, and flavor comments for the evaluated product, with missing comments inserted as `(empty)`. The source comments remained in French. The observed liking score was withheld from the scoring prompt and used only later as the prediction target. The fixed instruction told the model to act as a sensory and consumer scientist, focus only on sensory descriptions of the ham itself, ignore non-sensory comments, and return JSON only, using the schema `{"visual": X, "texture": X, "flavor": X}` with the active scale. The verbatim prompt-building function is given in A. Figure 1 summarizes the workflow.

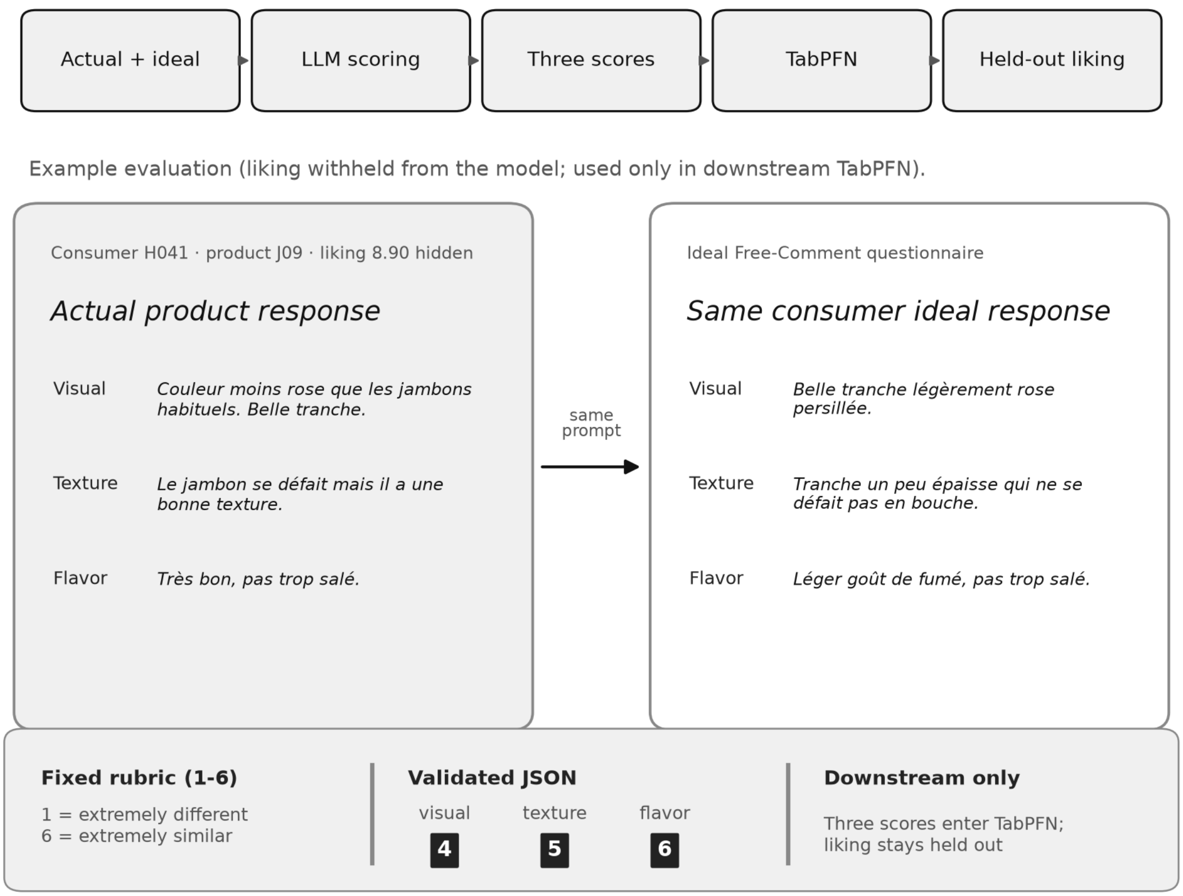

Figure 1: Direct LLM alignment scoring. Top: pipeline from paired Free-Comment text to held-out liking. Bottom: example evaluation (liking withheld from the model) with fixed 1-6 rubric, validated JSON scores, and TabPFN inputs.

2.2.4 Response parsing and scoring reliability

Evaluations were scored and validated independently. The parser first attempted direct JSON parsing, then used a regular-expression fallback for the three expected fields. A response was accepted only if all three fields were present and each score was an integer within the active scale range. For thinking-enabled responses, any output parts flagged as model thought text were skipped before parsing the answer. Each evaluation was attempted up to three times at the API (Section 2.2.1), not as local re-parses of a single response body.

No evaluation was dropped from modeling. If no valid response was obtained within the retry budget, the three modality scores for that evaluation were set to the scale midpoint, `(scale_points + 1) // 2`, that is, the lower of the two middle categories (2, 3, or 4 on the 4-, 6-, and 8-point scales), so all 2,758 evaluations entered every downstream model. We report a scoring-error count per configuration as a reliability indicator for the generation and parsing loop: an evaluation was flagged if an API error, exception, or invalid response occurred during its attempts, even when a later retry recovered a valid score. In plain terms, the error count measures how often a configuration failed to return a usable structured score on the first try. It is a measure of operational reliability under a fixed schema, not a count of excluded data and not a formal measure of task difficulty.

Configurations were also compared on operational criteria: wall-clock time for a full scoring pass over all 2,758 evaluations, the scoring-error rate defined above, and approximate API cost. We report cost qualitatively because provider prices change; for scale, a full direct-response scoring pass completed in about 100 seconds of wall-clock time at a cloud API cost that was small relative to analyst time.

2.3 Predicting liking from alignment scores

2.3.1 Primary learner

For the primary predictive evaluation, only the three modality alignment scores were used as predictors of 0–10 liking. The primary downstream learner was TabPFN, a tabular foundation model trained on more than 100 million synthetic tabular tasks generated from structural causal model priors [Hollmann et al., 2025]. TabPFN predicts on a new tabular dataset through in-context inference rather than ordinary hyperparameter search, which suits this workflow: the LLM produces compact tabular features, and TabPFN predicts liking without the tuning burden that would complicate a 27-configuration grid search with bootstrap validation. Liking prediction used TabPFN v2 via `TabPFNRegressor` [Hollmann et al., 2025], with the device set to CUDA when available and CPU otherwise and all other inference settings left at library defaults. No separate hyperparameter search was performed.

2.3.2 Validation design and metrics

The validation design used a single consumer-group holdout split with `GroupShuffleSplit(test_size = 0.20, n_splits = 1, random_state = 42)`, grouping by consumer. This prevented evaluations from the same consumer from appearing in both training and holdout sets. A bootstrap sensitivity analysis [Efron, 1979] was run on the two best single-split configurations from each model family, six configurations in all. It held the LLM scores fixed, rescaled each modality score to a common 0–10 range, and averaged the three values into a single alignment predictor. For each of 200 replicates, consumers were resampled with replacement, a fresh 20% consumer-group holdout was drawn within the resampled data, and TabPFN was refit. The bootstrap therefore quantifies sensitivity to the consumer sample and split. It does not repeat the LLM scoring, and because it uses the averaged single predictor it is reported separately from the primary three-feature result. Family-level win probabilities pooled the 200 R^2 draws from each family's two selected configurations and compared aligned draws.

Predictive performance is reported with the coefficient of determination (R^2) and the mean absolute error (MAE), computed over held-out evaluations:

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}, \qquad \mathrm{MAE} = \frac{1}{n} \sum_i |y_i - \hat{y}_i|,$$

where i indexes held-out consumer-product evaluations, y_i is the observed liking for evaluation i, $\hat{y}_i$ is the predicted liking, and $\bar{y}$ is the mean observed liking. R^2 measures the share of liking variance the model explains (accuracy), and MAE measures the average prediction error in points on the 0–10 liking scale. Both metrics pool the held-out evaluations; they are not computed on product means.

To compare the contributions of the three modality scores, we computed permutation importance on the held-out evaluations (`n_repeats = 10`, `random_state = 42`), averaged the importances across the 27 grid configurations, and normalized them within method for comparison with the published modality signal.

2.3.3 Downstream learner sensitivity analysis

As a sensitivity analysis, the same direct LLM scores were also evaluated with Ridge Regression [Hoerl and Kennard, 1970], Random Forest [Breiman, 2001], Gradient Boosting [Friedman, 2001], and XGBoost [Chen and Guestrin, 2016]. For that comparison, scores were rescaled to a common 0–10 range before fitting:

$$\frac{\text{score} - 1}{\text{scale\_points} - 1} \times 10.$$

The conventional learners used fixed, off-the-shelf settings rather than a nested hyperparameter search: Ridge `alpha = 1.0`; Random Forest `n_estimators = 100`, `max_depth = 10`, `random_state = 42`; Gradient Boosting `n_estimators = 100`, `max_depth = 5`, `random_state = 42`; and XGBoost `n_estimators = 100`, `max_depth = 5`, `random_state = 42`. This design keeps the comparison focused on the LLM scoring configuration rather than on separately optimized learner families. A fully tuned comparison would require nested tuning within the training portion of each consumer-group split.

2.4 Generic embedding-similarity baseline

As a reference floor, we also predicted liking from generic embedding similarity. This baseline converted text into vectors and measured geometric closeness without asking any model for a sensory judgment. In the primary analysis it used the provider `models/text-embedding-004` embedding model (768-dimensional vectors) through the REST `batchEmbedContents` endpoint with `taskType = "SEMANTIC_SIMILARITY"`. Comments were first reduced to modality-specific sensory descriptors by a separate LLM extraction step (`gemini-2.0-flash`) that retained only sensory terms for the target modality; actual and ideal descriptor texts were embedded separately for each modality; and cosine similarity was computed between the actual vector a and ideal vector i for each modality:

$$\cos(a, i) = \frac{a \cdot i}{\|a\|\|i\|}.$$

The three cosine similarities then entered the same downstream prediction as the alignment scores. This baseline did not enforce the visual/texture/flavor alignment judgment and did not receive liking. It is included as a simple reference floor, not as a tuned competitor. The reported $R^2 = 0.200$

is from that text-embedding-004 run (documented in the primary analysis notebook). The public archive also includes a separate EmbeddingGemma sensitivity experiment with different features and results; those files are not the source of the 0.200 reference figure.

2.5 Topic-level validation

The primary pipeline uses three alignment scores. To compare more directly with the Mahieu et al. [2022] descriptor analysis, we also ran a topic-level version with 17 topic-alignment scores grounded in the descriptor families from Mahieu et al. [2022]. Topics covered overall visual match, color and pinkness, fat and lean appearance, surface moisture and shine, slice structure, overall texture, tenderness, firmness and rubberiness, dryness and juiciness, fibrous pieces, thickness and chew, overall flavor, saltiness, ham taste, smoky or spiced notes, blandness or insipid intensity, and off-notes or aftertaste.

The topic-level run used the direct-response configuration (`gemini-flash-lite-latest`), temperature 0.7, and a six-point scale. Each evaluation was scored once on the 17 topics; the prompt allowed `null` when neither the actual nor the ideal text gave enough evidence for a topic. Of 2,758 source evaluations, 2,757 produced valid topic-level scores; the single evaluation that failed to parse was excluded from topic summaries.

Each topic score measures actual-versus-ideal alignment for that topic; these are not descriptor citation counts. Evaluation-level topic correlations used the consumer-product evaluation as the statistical unit and computed Spearman correlations between each topic score and that evaluation's liking. Product-level summaries averaged each topic score by product across consumers and correlated those 30 product means with product mean liking. We did not attempt to rerun the original MR-CA: the open dataset provides the raw comments and liking scores but not the curated descriptor citation matrix of Mahieu et al. [2022], and reconstructing that matrix would reintroduce the manual text-processing pipeline this approach is meant to avoid. Instead, we compared structure: the modality order and the rank agreement of matched topic families. Topic-rank agreement compared the strengths of the Mahieu et al. [2022] drivers, taken as absolute values and normalized to sum to one across matched families, with the LLM topic-liking correlations transformed the same way. We compare strength rather than sign because the two quantities have different meanings: a descriptor-presence effect can be negative when an undesirable descriptor is cited, while an actual-to-ideal alignment score can be positive when an undesirable attribute is absent. Rank agreement is summarized with Spearman's ρ and Kendall's τ_b, a rank correlation that handles ties [Kendall, 1938], with a two-sided permutation test for τ_b.

For a topic-level predictive check, the 17 topic scores were used to predict liking under the same consumer-group holdout. The full topic-level TabPFN fit exceeded the memory available in the compute environment used for that run, so this check used a Gradient Boosting regressor [Friedman, 2001] with scikit-learn default settings (100 trees, maximum depth 3) and `random_state = 42` as a diagnostic fallback. It is an extension of the structural validation, not a replacement for the primary three-feature TabPFN result.

Observed liking is treated throughout as the criterion variable, not as a perfect ground truth for consumer preference. The study asks whether LLM-derived actual-to-ideal alignment scores predict stated liking and recover the published sensory structure; it does not assume that every liking response is noise-free.

2.6 Validation against held-out Just-About-Right ratings

Recovering the published modality and topic structure is reassuring, but liking is part of the same study the structure came from, so agreement could partly reflect a shared hedonic signal. The dataset also contains a separate structured signal: the JAR ratings for color, fat, salt, and tenderness (Section 2.1). These were collected as distinct questions, were never included in any scoring prompt, and map onto four of the topic-alignment scores (color and pinkness, fat and lean appearance, saltiness, and tenderness). They therefore allow a prompt-held-out construct check from the same study sessions: if the model read the open comments correctly, its alignment score for an attribute should be highest when the consumer rated that attribute just about right and should fall as the rating moves toward either extreme. This is internal validation against ratings not shown to the model, not an external dataset replication.

For each of the four attributes we computed mean LLM alignment by JAR level, the Spearman correlation between alignment and the absolute JAR deviation |JAR|, and the signed correlation with JAR. To test attribute specificity rather than a general hedonic halo, we also correlated every LLM attribute score with every JAR attribute, giving a four-by-four matrix in which a diagonal pattern indicates attribute-specific recovery. The primary validation used the same direct-response six-point topic-level scores as Section 2.5, joined to the JAR ratings on consumer and product.

To ask whether thinking improves this recovery, we repeated the topic-level scoring with the two thinking configurations from the main grid (minimal and low thinking) at the same six-point scale and temperature 0.7, and recomputed the same recovery and specificity metrics for each configuration. We also ran a prompt-sensitivity check on a fixed 250-evaluation sample, scoring all three configurations under the baseline prompt and under a more explicit rubric prompt written to favor deliberative scoring, to test whether the order of configurations on JAR recovery was stable across prompts.

2.7 Software and implementation

Analyses were implemented in Python 3.9.6. The embedding-baseline and TabPFN environment recorded pandas 2.3.3, scikit-learn 1.6.1, TabPFN (tabpfn 7.1.1; ModelVersion.V2), and PyTorch 2.8.0. Downstream learner comparisons also used XGBoost via `XGBRegressor` where available; the historical XGBoost package version was not recorded in the archived environment metadata, so that secondary comparison is not exactly environment-reproducible. The primary grid liking models used TabPFN library defaults with device set to CUDA or CPU. LLM scoring and text embedding used a commercial API from one provider (Google Gemini when we ran the work). Scoring scripts, configuration-level grid summaries, bootstrap outputs, topic-level per-evaluation alignment scores, and figure generators are archived in the public repository described in the Data and code availability statement.

3 Results

3.1 Predictive performance across the scoring grid

The best of the 27 grid configurations used the direct-response setting, a six-point scale, and temperature 0.7. It reached $R^2 = 0.573$ and MAE = 1.22 on the main consumer-group holdout split. The bootstrap sensitivity analysis for this configuration produced mean $R^2 = 0.504$ with a 95% interval of [0.429, 0.582]; as described in Section 2.3.2, that analysis uses the mean of the three normalized alignment scores as a single predictor, so it is reported separately from the primary three-feature estimate. To avoid confounding thinking style with scale size or temperature, Table 3 compares the three configurations at the same six-point scale and temperature 0.7.

Table 3: Controlled predictive and operational comparison at six points and temperature 0.7. Time is wall-clock time for a full scoring pass over 2,758 evaluations; scoring errors are evaluations with at least one failed generation or parse attempt (Section 2.2.4).

Configuration	Scale, temp.	R^2	MAE	Time (s)	Scoring errors
Embedding baseline	–	0.200	1.83	<5	0 (0.0%)
Direct-response	6 pt, 0.7	0.573	1.22	101	114 (4.1%)
Low thinking	6 pt, 0.7	0.479	1.33	619	137 (5.0%)
Minimal thinking	6 pt, 0.7	0.451	1.36	133	209 (7.6%)

The full grid supports the same conclusion without relying on one scale setting (Figure 2). Direct-response cells had the highest mean R^2 for the direct-response group and the best overall cell, and all nine direct-response cells occupied nine of the top ten positions by R^2. The best low-thinking cell used an eight-point scale and temperature 0.7, with $R^2 = 0.518$ and $\text{MAE} = 1.26$, still below the best direct-response result; the best minimal-thinking cell reached $R^2 = 0.465$. Bootstrap comparisons among selected top cells favored the direct-response class, with estimated winning probabilities of 99.3% versus minimal thinking and 91.2% versus low thinking (Figure 2). These are descriptive bootstrap comparisons of the selected configurations, not multiplicity-adjusted tests of all 27 cells, and not a claim that every future model will order the same way. Full per-configuration results for all 27 setups are given in B and in the project archive.

Within the thinking-capable generation, more thinking did not behave as a simple improvement. Low thinking was not worse than minimal thinking on prediction; at the controlled setting it was somewhat better (Table 3). Neither thinking level closed the gap to the direct-response configuration.

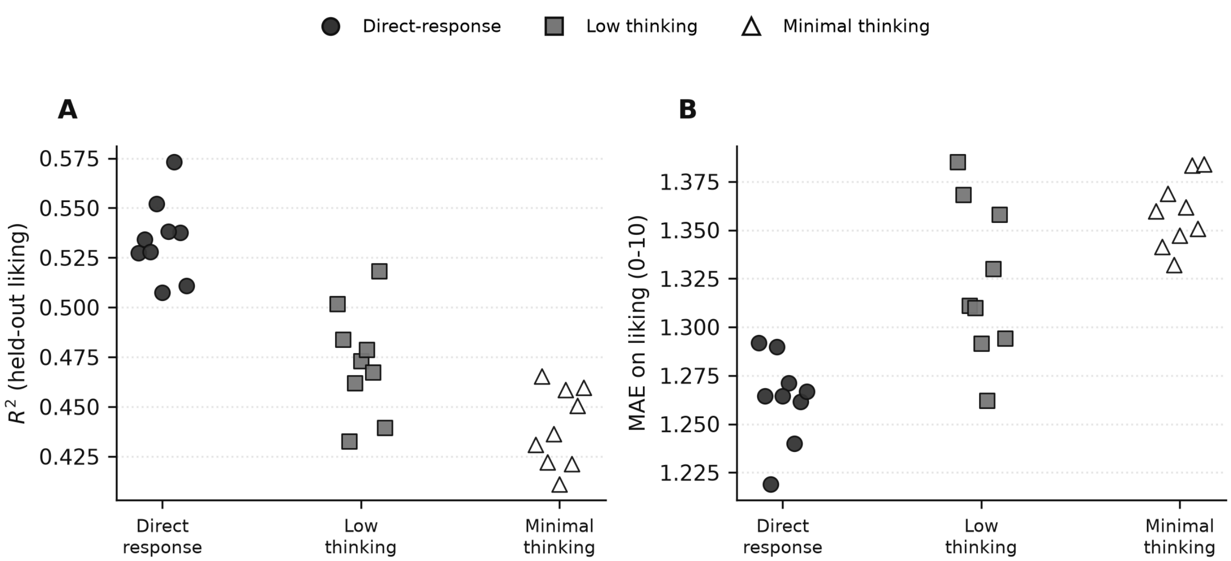

Figure 2: Performance across the 27-configuration grid. Points are individual scale-by-temperature cells for each model configuration. (A) R^2 on held-out liking. (B) MAE on liking (0-10).

3.2 Operational comparison

The thinking configurations also carried operational costs (Table 3 and Figure 3). Wall-clock time for a full scoring pass rose from 101 seconds for the direct-response configuration to 619 seconds at low thinking. At the same six-point scale and temperature 0.7, minimal thinking produced the most scoring errors (209 evaluations, 7.6%, versus 114, 4.1%, for direct-response). These failures matter because the pipeline depends on thousands of parseable JSON responses; as noted in Section 2.2.4, they measure operational reliability under a fixed schema rather than task difficulty.

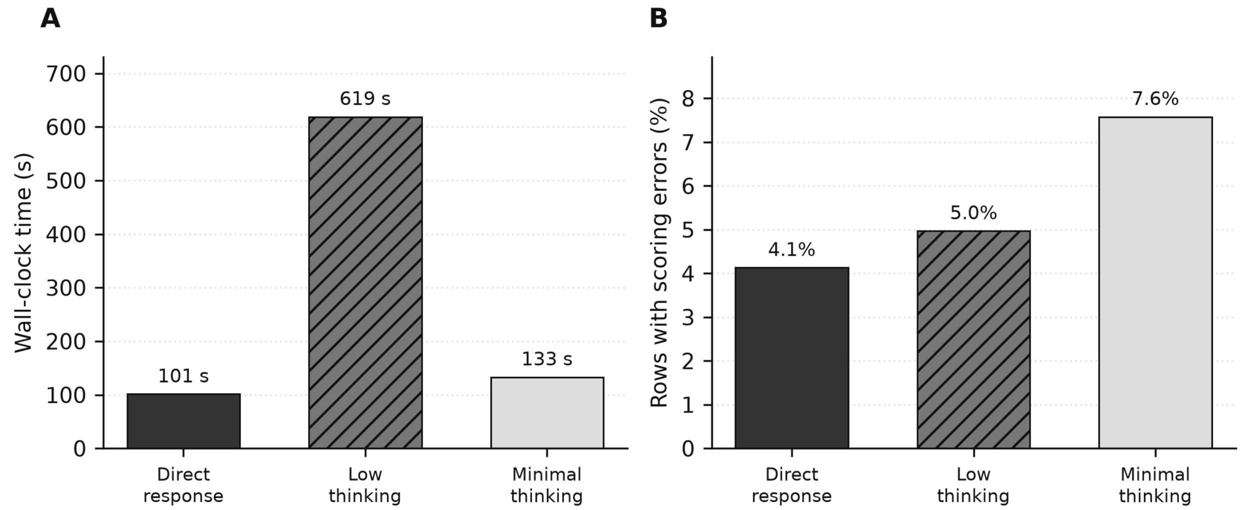

Figure 3: Operational comparison for the controlled six-point, temperature 0.7 configurations. A useful scorer must be accurate, fast enough to run across thousands of evaluations, and reliable enough to return parseable structured output.

3.3 Comparison with the embedding baseline

The generic embedding baseline reached $R^2 = 0.200$ and MAE = 1.83, well below the $R^2 = 0.573$ of the best alignment-scoring configuration (Table 3). This gap is informative because the task is more than deciding whether two comments are semantically close. A consumer might describe an actual ham and an ideal ham with overlapping words but different sensory implications, and another consumer might use different words to describe a close match. Raw cosine similarity does not force the comparison to respect modality, hedonic direction, or consumer-specific ideals.

3.4 Downstream learner comparison

The downstream learner comparison matters for the workflow rather than for the headline result (Table 4). Across the 27 LLM scoring configurations, TabPFN had the best mean R^2 at 0.482 and won 20 of 27 configurations; Ridge Regression was the closest conventional baseline at 0.468. The conventional learners were untuned (Section 2.3.3), so this comparison shows that TabPFN is competitive and operationally convenient, not that it is uniformly the best possible learner. The margin was modest, but TabPFN required no separate hyperparameter search and runs as a single tabular forward pass, which kept the 27-configuration grid and bootstrap loop practical.

Table 4: Downstream learner comparison across the 27 LLM scoring configurations. Conventional learners used fixed, untuned settings.

Downstream learner	Mean R^2	Configurations won
TabPFN v2	0.482	20 (74.1%)
Ridge Regression	0.468	7 (25.9%)
Gradient Boosting	0.450	0
Random Forest	0.446	0
XGBoost	0.438	0

3.5 Modality hierarchy and topic-level structure

Permutation importance from the three-score model (Section 2.3.2) gave the largest role to flavor, a smaller role to texture, and the weakest role to visual appearance (Figure 4). This matches the broad pattern in Mahieu et al. [2022].

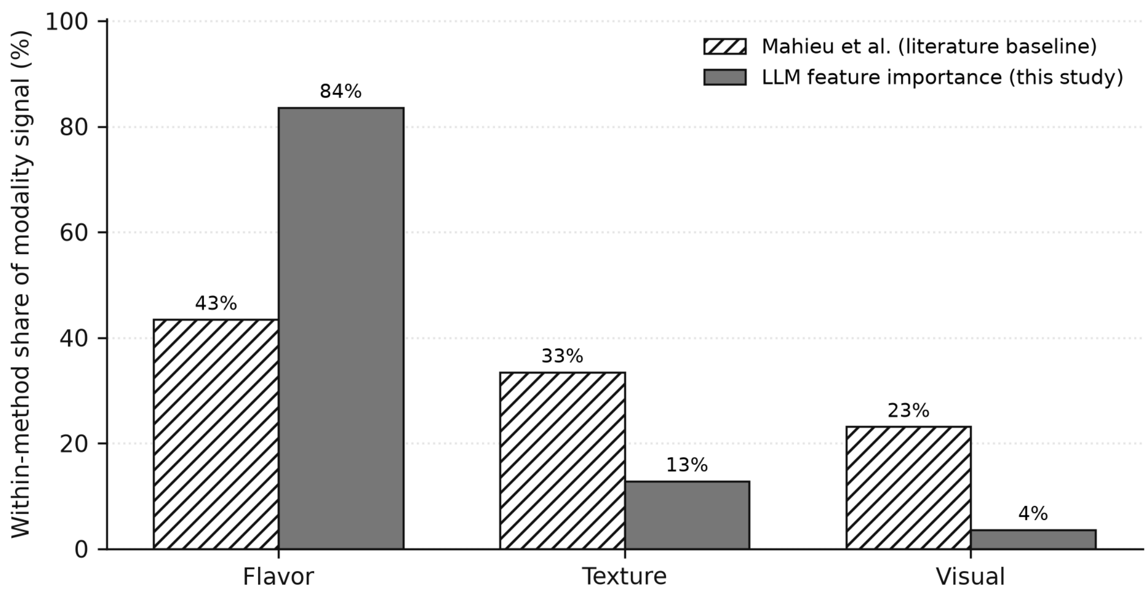

Figure 4: Modality-level sanity check. Values are normalized within method so the comparison is about order and relative share, not raw units. Mahieu et al. [2022] values summarize modality signal from the original descriptor-based analysis, while our values are held-out permutation importances from the three LLM alignment scores, averaged across the 27 configurations.

The topic-level analysis strengthened that result (Table 5). It recovered the same modality order and showed substantial rank agreement across matched topic families. Across 17 matched topic families, Kendall's τ_b was 0.519 with a two-sided permutation $p = 0.003$, Spearman's ρ was 0.715, and nine of the ten strongest topic families in the original analysis were also among the ten strongest LLM topic families (Figure 5). The topic-level predictive check with the Gradient Boosting fallback reached $R^2 = 0.470$ and MAE = 1.37.

Table 5: Topic-level validation results (direct-response configuration, six-point scale, temperature 0.7).

Quantity	Result
Valid topic-level evaluations	2,757
Unparsed evaluations	1
Topic families	17
Original modality order	Flavor > Texture > Visual
LLM modality order	Flavor > Texture > Visual
Modality Spearman ρ	1.000
Topic rank Spearman ρ	0.715
Topic rank Kendall τ_b	0.519
Topic-score liking prediction (GB fallback)	$R^2 = 0.470$; MAE = 1.37

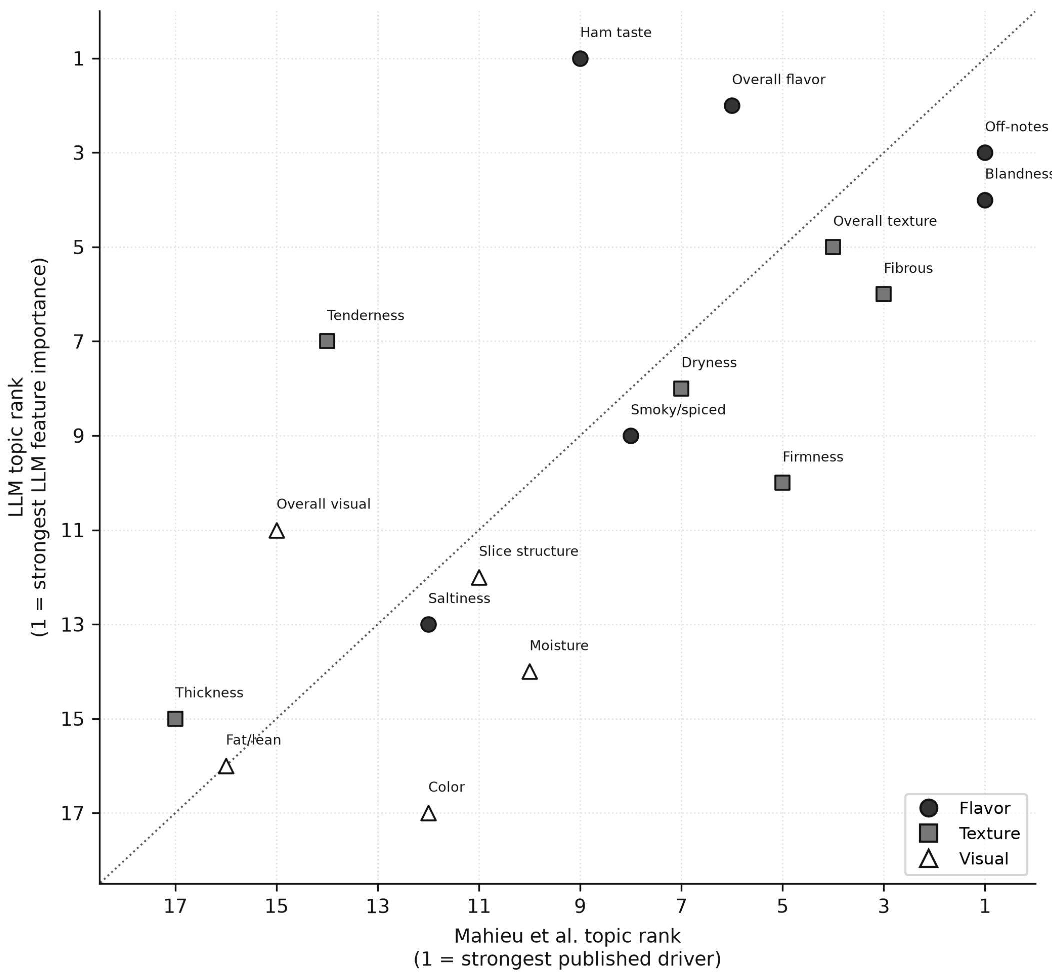

Figure 5: Rank-order comparison between mapped Mahieu et al. [2022] descriptor families and LLM topic-alignment families. Lower ranks are stronger. The comparison covers 17 matched topic families, with Kendall's $\tau_b = 0.519$ and Spearman's $\rho = 0.715$.

3.6 Recovery of held-out Just-About-Right ratings

The modality and topic-rank results above compare the LLM scores with structure that was itself derived from liking. A stronger test uses the held-out JAR ratings, which the model never saw. For all four attributes the LLM alignment score peaked when the consumer rated the attribute just about right and declined toward both the "not enough" and "too much" extremes, the classic JAR tent shape (Figure 6). The correlation between alignment and absolute JAR deviation was negative for every attribute: saltiness $\rho = -0.68$ (n = 1,662), tenderness $\rho = -0.59$ (n = 1,878), fat and lean $\rho = -0.50$ (n = 1,635), and color $\rho = -0.46$ (n = 1,691), all with nominal evaluation-level $p < 10^{-10}$. The model recovered these structured human judgments from free text it was only asked to compare against each consumer's ideal.

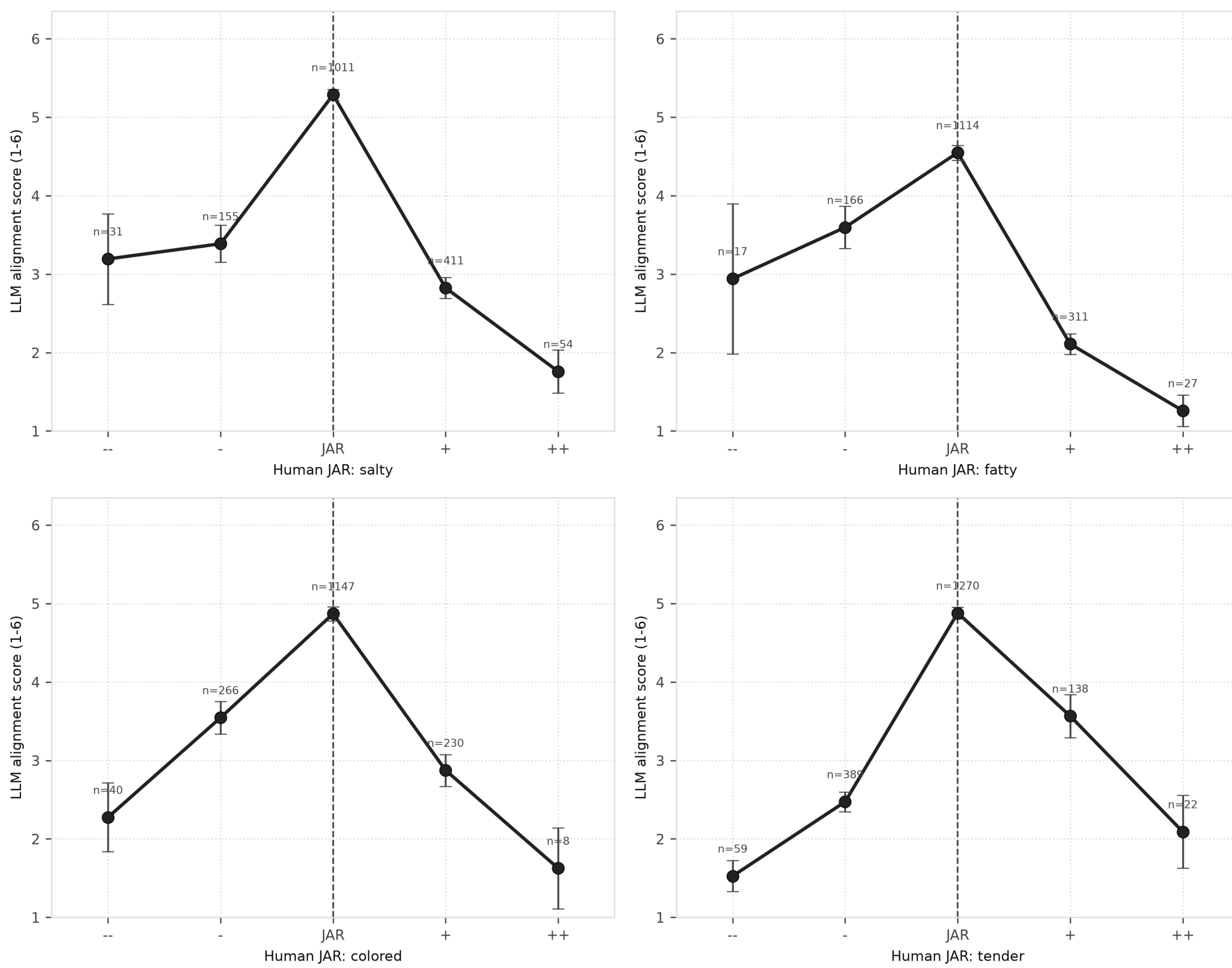

Figure 6: LLM free-text alignment scores recover the human Just-About-Right structure. Each panel plots mean LLM alignment (1–6 similarity to the consumer's ideal) against the consumer's JAR rating for that attribute, from −2 (not enough) through 0 (just about right) to +2 (too much), with 95% confidence intervals. The JAR ratings were never shown to the model.

The recovery was attribute-specific rather than a hedonic halo. Each LLM attribute score tracked its own JAR rating more strongly than any other (Table 6): the mean matched (diagonal) correlation was −0.555 against a mean mismatched correlation of −0.251, a specificity gap of 0.30, and for every attribute the strongest recovery was its own JAR.

Table 6: Attribute specificity. Spearman ρ between each LLM attribute alignment score (rows) and each absolute JAR deviation (columns). More negative is stronger recovery; the diagonal is strongest in every row.

LLM score	|JAR salt|	|JAR fat|	|JAR color|	|JAR tender|
Saltiness	**-0.68**	-0.23	-0.25	-0.24
Fat / lean	-0.24	**-0.50**	-0.35	-0.26
Color	-0.20	-0.23	**-0.45**	-0.20
Tenderness	-0.23	-0.30	-0.27	**-0.59**

The signed correlations recovered the deviation directions that matter for cooked ham, in agreement with the drivers in Mahieu et al. [2022]. Alignment fell faster on the "too salty" side than on the "not salty enough" side (signed $\rho = -0.37$), and likewise for "too fatty" (signed $\rho = -0.34$). For tenderness the penalized direction was "not tender enough" (signed $\rho = +0.40$). Color was symmetric (signed $\rho \approx 0$): both too pale and too dark reduced alignment. The model therefore recovered which deviation direction lowers the match to the ideal for each attribute, in addition to the overall tent shape.

Thinking did not improve this recovery. Scoring the same evaluations with the two thinking configurations gave essentially the same result: mean matched recovery was 0.555 for the direct-response configuration, 0.560 for minimal thinking, and 0.548 for low thinking, with comparable specificity gaps (0.30 to 0.33) and comparable data completeness (Figure 7). The per-attribute correlations differed by at most a few hundredths across configurations. The prompt-sensitivity check on a fixed 250-evaluation sample was consistent with a near tie: minimal thinking ranked slightly first under both prompts, while direct response and low thinking exchanged second and third place, and each configuration's recovery moved by less than 0.02 between the baseline and rubric prompts. This is descriptive agreement, not a formal equivalence test. The thinking configurations were also far slower, with low thinking taking several times the direct-response runtime. The direct-response configuration therefore matched the thinking configurations at reading structured sensory judgments out of free text; its advantage in the main analysis was specific to liking prediction and to speed and reliability, not to attribute perception.

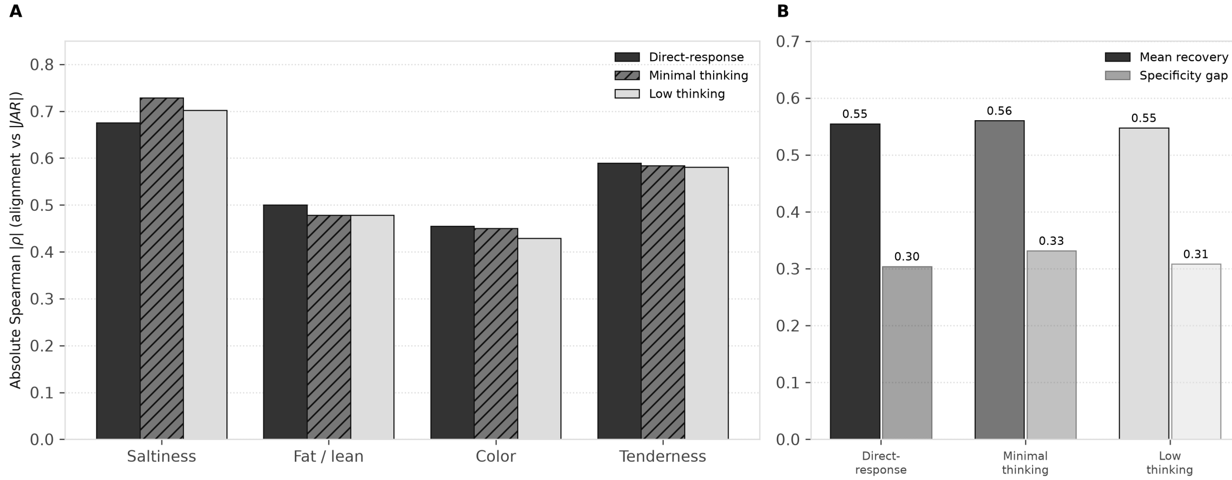

Figure 7: Cross-configuration JAR recovery. (A) Absolute Spearman correlation of alignment with |JAR| by attribute. (B) Mean matched recovery and specificity gap. Thinking configurations do not improve recovery relative to the direct-response configuration.

3.7 Illustrative evaluation-level disagreements

To make the aggregate contrast concrete, we inspected evaluations where the configurations disagreed. In one high-liking case, consumer H041 gave the product a liking score of 8.90. The actual comments included “The ham breaks apart, but it has a good texture” and “Very good, not too salty.” The ideal comments asked for a slightly thick slice that does not fall apart in the mouth, is not too dry, and has a light smoky taste that is not too salty.

The two configurations received the same evaluation, prompt, and 1 to 6 similarity scale. The direct-response configuration scored the evaluation 4, 5, and 6 for visual, texture, and flavor similarity to the ideal. The low-thinking configuration scored it 4, 2, and 4. A plausible interpretation is that the thinking configuration read the mention of the ham breaking apart as a severe literal mismatch with the ideal slice description, while the direct-response configuration weighted the consumer’s overall positive sensory description. Such examples do not replace the statistical results. They illustrate how a more deliberative setup may over-read local mismatches in consumer language (how the model scored the text, not the JSON parsing described in Section 2.2.4) and convert them into severe similarity penalties even when liking is high.

4 Discussion

We began this work expecting the more deliberative configurations to be better at this scoring job. The data pointed the other way. Among the setups we tested, the best sensory scorer was the fast, direct-response configuration: it predicted held-out liking better, ran roughly six times faster than the low-thinking configuration, and returned parseable structured output more reliably (Section 3.1 and Section 3.2). That reversal fits the nature of the target response. Liking a ham is a fast, sensory, affective judgment. The scoring prompt asks the model to judge whether the actual product matched the consumer’s ideal experience, not to solve a multi-step reasoning problem. In the dual-process analogy of Section 1, the task looks like a fast judgment, and the setup matched to that regime measured it best. We treat this mapping as a hypothesis for sensometrics, not as evidence about how models work inside. If sensory and consumer tasks systematically favor different styles of inference, model-cognition fit becomes a research program of its own for the field.

The comparison also gives a within-model reading. The two thinking configurations differ only in thinking level, and moving from minimal to low thinking did not behave as a simple improvement: low thinking was somewhat better on prediction, the two were essentially tied on JAR recovery, and neither closed the gap to the direct-response configuration (Section 3.1 and Section 3.6). We do not conclude that deeper thinking is generally better or generally worse. We conclude that the thinking setting changed the measurement, in cost and in quality, and that nothing about the more deliberative options made them automatically superior. That is the finding a researcher should carry into their own workflow: model settings matter, and the direction of their effect on a given task is an empirical question. Because LLM workflows are often judged as if capability were one-dimensional, the natural default is to reach for the most advanced or most deliberative option. On this task that default would have produced a slower, less accurate, less reliable scorer.

Beyond the configuration contrast, direct LLM alignment scoring turned IFC data into useful predictors of liking. The model received paired actual and ideal consumer comments, scored sensory alignment, and produced tabular features that predicted liking far better than generic embedding similarity ($R^2 = 0.573$ versus 0.200, Section 3.3) and recovered the same broad sensory hierarchy as the original descriptor-based analysis (Section 3.5). This complements the wider move toward machine-learning prediction of liking from sensory data, such as the recent prediction of coffee liking from sensory and JAR features [Gunning et al., 2026]; the contribution here is that the predictors come directly from open comments rather than from a pre-structured sensory table. The approach sits between classical sensory statistics and newer LLM scoring methods. It keeps the sensory structure of the problem, since the prompt asks for visual, texture, and flavor comparisons and the topic-level extension uses families grounded in the original descriptor analysis, while avoiding much of the manual work of cleaning, lemmatizing, grouping, and filtering free comments before modeling.

The method does change what is measured. Descriptor-presence analysis asks what terms were cited and how their presence relates to liking. Alignment scoring asks whether the actual product matched the consumer's ideal. These are related but not identical targets, and signed descriptor coefficients should not be compared one-to-one with our feature importances. The appropriate comparison is broader structure: modality order and topic-rank agreement, which is what Section 3.5 evaluates. For liking prediction, the relational actual-versus-ideal score may be closer to the consumer decision process than descriptor presence alone.

A natural concern is circularity: would a model that simply read the hedonic tone of the actual comment track liking trivially? Three observations argue against this. First, comparing each actual product against the consumer's own ideal is the defining construct of IFC and of IPM [Mahieu et al., 2022, Worch et al., 2013], not a quirk of the present workflow. Mahieu et al. [2022] showed that ideals genuinely differ across consumers, separating the panel into segments with opposite flavor ideals; a score that merely echoed the actual comment could not exploit that consumer-specific reference. Second, the recovery of the held-out JAR ratings was attribute-specific (Section 3.6): each alignment score tracked its own JAR attribute far more strongly than the others, which a generic hedonic or sentiment signal would not produce. Third, the generic embedding baseline operated on the same actual and ideal text but reached only $R^2 = 0.200$, so the predictive signal does not follow from the text alone but from the structured actual-versus-ideal sensory judgment. Hedonic tone in the comments surely contributes signal; the evidence indicates it is not the whole story.

These results also bear on the tradeoff between speed, cost, and evidence quality that constrains consumer research. The original analysis required a manual expert pipeline of cleaning, lemmatization, descriptor grouping, frequency filtering, and cross-tabulation before any model could be fit, the kind of preprocessing whose automation is itself under study [Visalli et al., 2025a]. That pretreatment took an experienced analyst about half a day per sensory modality, roughly two days in total including overhead, to turn the raw comments into the descriptor table used for modeling. Direct alignment scoring of all 2,758 evaluations took about 100 seconds of wall-clock time with the direct-response configuration (Table 3), at a cloud API cost that was small relative to that analyst time. The tradeoff does not vanish; it moves. The recurring per-dataset cost of hand-coding descriptors is replaced by a one-time engineering cost of prompt design, calibration, and model versioning. The thinking-configuration results add a caution to that engineering: spending more model compute per evaluation is not where the measurement gains were found in this study.

The comparison should be treated as something that ages and needs rechecking. Named API models move on quickly relative to peer review. That fact does not weaken the result; it is why reported identifiers and scoring dates matter. Model releases move the baseline, so a task-matched pilot (Section 5) should be rerun after major model changes, and reported results should record the aliases, thinking settings, temperatures, and scoring dates used. Judge-model and prompt choices are themselves sources of measurement error in LLM workflows [Messing, 2026], which is a further reason to keep the scoring loop auditable.

What should travel beyond this dataset is not a fixed ranking of Gemini products. It is the demonstration that thinking level can matter, sometimes in a surprising direction, on a sensory free-text task, together with a model-cognition-fit account that invites further work. Other teams, using other open or closed models and other product categories, should run their own pilots rather than copy which setup ranked first for us. The analogy with fast versus deliberate human judgment is offered as a prompt for further work, not as a finished theory of how LLMs work.

5 Recommendations for practice

This paper does not recommend a specific commercial model. The commercial models we used will be superseded, and a recommendation to adopt any named system would be exactly the kind of blind default the results warn against. What the results support is a discipline: before locking an LLM scoring workflow for a full study, run a small task-matched pilot that varies thinking level among other settings.

1. Define the judgment the model must make and the human signal that validates it (here, actual-versus-ideal alignment validated against liking and held-out JAR ratings).

2. Set aside a small sample of evaluations, including short, missing, and contradictory comments, and hold everything constant except the model setup.

3. Compare setups, direct-response versus thinking levels, rather than brand stories, and include at least one setup of each style.

4. Rank setups on task criteria: validity against the human signal, structured-output reliability, runtime, and cost.

5. Prefer a prompt-held-out check the model never saw, as with the JAR ratings here, over reusing the training signal.

6. Lock the winning setup with static model identifiers and a scoring date, archive prompts and parsed scores, and re-check after major model releases.

The pilot cost is small relative to a full study: in this work a full scoring pass over 2,758 evaluations took about 100 seconds with the direct-response configuration, and a pilot needs only a fraction of that. The same discipline applies to open-weight and closed models alike. Choosing between them is a research-management tradeoff involving access, cost, privacy, latency, and reproducibility, and nothing in the fit idea is specific to closed systems. Open-weight models offer stronger reproducibility guarantees because the weights can be archived; the pilot loop above is how a team would establish whether a given open or closed setup measures their task well.

6 Limitations and future work

This is a single-product-category study. Cooked ham has a clear sensory structure, and flavor dominates liking in the original analysis. Other categories may have different modality balances and may not favor direct-response configurations to the same extent.

A central limitation is scope of models. We stayed in one family of commercial fast models (Gemini configurations when we ran the work) so that we could explore thinking versus direct response without mixing provider stacks. That design is useful for a controlled exploration; it is not a general survey of LLMs. Open-weight models, other closed providers, and later generations may order differently. Readers should treat our results as findings from this exploration, not as universal. The scientific claim we stand behind is that thinking level is worth testing on sensory text-scoring tasks, not that the specific ranking of settings will hold for every laboratory.

Within that family the comparison spans a direct-response setup and two thinking levels of a related generation. The direct-response configuration also differs from the two thinking configurations in generation, so the headline contrast cannot attribute the gap to the thinking setting alone; only the within-model contrast isolates thinking level. Higher thinking levels were not tested. Within the thinking-capable generation at six points and temperature 0.7, low thinking improved prediction and structured-output reliability relative to minimal thinking but increased runtime, and neither closed the gap to the direct-response configuration. Whether some task exists at which higher thinking levels would dominate is an open question.

Because providers update aliases, what lasts is the pilot loop and the saved scoring outputs. The public archive records the exact alias strings used, the thinking configuration, the temperature, and the scoring date for each run, together with configuration-level primary-grid summaries, topic-level per-evaluation scores, and scripts that regenerate the reported downstream tables from those fixed outputs. Rerunning live provider aliases is not expected to reproduce the 15 January 2026 responses exactly. Other researchers should run their own pilots on their own data rather than adopt our configuration labels as a default.

Direct LLM alignment scoring is not SSR. SSR may provide better-calibrated distributions and uncertainty information [Maier et al., 2025]. A fair comparison should ask whether SSR's probability distributions improve calibration or whether direct scoring wins because the prompt focuses the model on the sensory comparison of interest.

The topic-level predictive check uses Gradient Boosting rather than TabPFN because the full topic-level TabPFN fit exceeded available memory (Section 2.5). It is a diagnostic extension of the modality and topic-rank validation, not a replacement for the primary three-feature TabPFN result. The topic-level comparison to Mahieu et al. [2022] uses mapped descriptor families and driver strengths coded from the published figure and processed notes, and should be read as a structural validation of modality and topic order, not as a re-estimation of descriptor coefficients.

The JAR validation is an internal construct check rather than a fully external one. The JAR ratings were collected in the same study sessions as the comments and liking ratings, so they share method variance, and topic scores were null where the text gave no evidence for an attribute, leaving per-attribute samples of roughly 1,600 to 1,900 evaluations. The evaluation-level correlation tests also do not adjust for repeated consumers and products. Replicating the JAR recovery on an independent dataset is future work.

Headline prediction used a single consumer-group holdout after examining 27 configurations, and that search was not corrected for multiplicity. The bootstrap resamples consumers and splits for six selected configurations, holds the LLM scores fixed, and averages the three modality scores into one predictor, so it does not repeat LLM generation, nest the configuration selection, or provide an interval for the primary three-feature estimate. Each configuration was scored once, including the temperature 0.7 runs, so repeated-generation variability is not quantified, and the midpoint fallback assigns the lower of the two middle categories, which could bias the few affected evaluations slightly toward the different side. Repeated grouped cross-validation and a pre-specified primary contrast would further harden which setup ranks first, and a fully tuned comparison of downstream learners would require nested tuning inside each training split; here TabPFN is reported as competitive and convenient in practice rather than as the uniformly best learner.

Finally, a model-person fit question is worth pursuing: whether some consumers' comments are better scored by direct-response configurations and others by more deliberative ones.

7 Conclusion

Direct LLM alignment scoring is a practical way to analyze Ideal-Free-Comment data. In the cooked-ham case study it turned paired actual and ideal comments into compact predictors of liking, outperformed generic embedding similarity, recovered the main sensory hierarchy from the original Free-Comment analysis, and recovered held-out Just-About-Right ratings attribute by attribute. Within the Gemini family we used for this exploration, the best scorer among the setups we tested was the direct-response setting. Setups with thinking turned on were slower and did not close the predictive gap. That pattern held across the configurations we ran, but it is a finding from this exploration, not a general law. The point of the paper is that thinking level matters for sensory free-text analysis, and that it can matter in a surprising way: more deliberation was not automatically better. A simple comparison with how people make fast sensory judgments versus more deliberate reasoning offers one limited way to think about why that might be so; the human brain is far more complex than any LLM, yet the parallel may open useful lines of research for sensometricians. Other teams should try thinking settings on their own tasks and tools, open or closed, rather than copy which setup ranked first for us.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During preparation of this manuscript, the authors used AI-assisted tools to support literature synthesis, manuscript organization, drafting and review of analysis code, and language editing. The human authors wrote, reviewed, and edited the manuscript, take full responsibility for all scientific content, analyses, and conclusions, and approved the final wording. All figures were produced by the authors' own plotting code from the study's analysis outputs; no figures or data were created or altered by generative image models.

Data and code availability

Supporting materials for this paper (open consumer dataset used here, configuration-level primary-grid summaries and bootstrap tables, topic-level per-evaluation alignment scores, JAR validation outputs, analysis scripts, figure generators, the primary scoring notebook, and manuscript sources) are available at https://github.com/aigorahub/model-cognition-fit-supporting-materials. That repository exists only to support this manuscript. It is not a general software product and does not include peer-review correspondence, credentials, raw API response bodies, or private computing infrastructure. Per-evaluation scores for the full 27-configuration primary grid are not included, so that grid is auditable at the configuration-summary level; topic-level analyses ship per-evaluation scores. A separate EmbeddingGemma sensitivity folder is not the source of the manuscript's text-embedding-004 $R^2 = 0.200$ floor. Full re-scoring with a live model API is optional and requires the reader's own credentials; the archived tables and scoring date are the durable audit trail for the reported results.

A Verbatim scoring prompt

The exact prompt template used for all direct-scoring configurations was:

```
def build_prompt(row, scale_points, scale_labels):
    """Build prompt with dynamic scale"""

    scale_text = "\n".join([f"  {i+1} = {label}" for i, label in enumerate(scale_labels)])

    prompt = f"""
You are a sensory and consumer scientist conducting consumer research on ham products.

TASK: Compare a consumer's ACTUAL ham experience to their IDEAL ham experience.
Score the alignment on each sensory dimension.

SCALE (1-{scale_points}):
{scale_text}

Focus only on sensory descriptions of the ham itself. Ignore any non-sensory comments.
Note: Responses are from French consumers and written in French.

IDEAL EXPERIENCE:
- Visual: {row.get('IdealVisual', '') or '(empty)'}
- Texture: {row.get('IdealTexture', '') or '(empty)'}
- Flavor: {row.get('IdealFlavor', '') or '(empty)'}

ACTUAL EXPERIENCE:
- Visual: {row.get('DescriptionVisual', '') or '(empty)'}
- Texture: {row.get('DescriptionTexture', '') or '(empty)'}
- Flavor: {row.get('DescriptionFlavor', '') or '(empty)'}

Return JSON only: {{"visual": X, "texture": X, "flavor": X}}
"""
    return prompt.strip()
```

The prompt inserted one of three ordered scales: a four-point scale (Very Different; Somewhat Different; Somewhat Similar; Very Similar), a six-point scale (Extremely Different; Very Different; Somewhat Different; Somewhat Similar; Very Similar; Extremely Similar), or an eight-point scale (Extremely Different; Very Different; Moderately Different; Slightly Different; Slightly Similar;

Moderately Similar; Very Similar; Extremely Similar).

B Full 27-configuration grid

Table 7: Primary split results for all 27 direct alignment configurations. Configuration labels use the archive aliases for auditability; the scientific contrast is direct-response versus thinking level within one family of fast models at the time of scoring. Time is wall-clock seconds for a full scoring pass over 2,758 evaluations; errors are evaluations with at least one failed generation or parse attempt (Section 2.2.4).

Configuration	Scale	Temp.	R^2	MAE	Time (s)	Errors
Gemini 2.5 Flash Lite direct response	4	0.0	0.5108	1.2919	102.3	118
Gemini 2.5 Flash Lite direct response	4	0.3	0.5274	1.2711	101.1	121
Gemini 2.5 Flash Lite direct response	4	0.7	0.5377	1.2644	101.2	126
Gemini 2.5 Flash Lite direct response	6	0.0	0.5342	1.2643	101.3	104
Gemini 2.5 Flash Lite direct response	6	0.3	0.5521	1.2399	101.3	112
Gemini 2.5 Flash Lite direct response	6	0.7	0.5731	1.2190	101.3	114
Gemini 2.5 Flash Lite direct response	8	0.0	0.5279	1.2616	103.9	109
Gemini 2.5 Flash Lite direct response	8	0.3	0.5075	1.2899	101.3	112
Gemini 2.5 Flash Lite direct response	8	0.7	0.5383	1.2669	100.6	114
Gemini 3 Flash minimal thinking	4	0.0	0.4214	1.3834	123.7	113
Gemini 3 Flash minimal thinking	4	0.3	0.4364	1.3689	124.9	125
Gemini 3 Flash minimal thinking	4	0.7	0.4111	1.3839	126.9	131
Gemini 3 Flash minimal thinking	6	0.0	0.4597	1.3507	137.3	196
Gemini 3 Flash minimal thinking	6	0.3	0.4654	1.3320	134.0	204
Gemini 3 Flash minimal thinking	6	0.7	0.4506	1.3618	133.0	209
Gemini 3 Flash minimal thinking	8	0.0	0.4311	1.3474	133.5	229
Gemini 3 Flash minimal thinking	8	0.3	0.4222	1.3599	134.1	232
Gemini 3 Flash minimal thinking	8	0.7	0.4586	1.3413	132.7	231
Gemini 3 Flash low thinking	4	0.0	0.4619	1.3580	547.8	85
Gemini 3 Flash low thinking	4	0.3	0.4327	1.3851	553.8	89
Gemini 3 Flash low thinking	4	0.7	0.4394	1.3683	582.6	87
Gemini 3 Flash low thinking	6	0.0	0.4730	1.3111	615.2	134
Gemini 3 Flash low thinking	6	0.3	0.4840	1.2917	614.6	128
Gemini 3 Flash low thinking	6	0.7	0.4787	1.3301	618.9	137

Configuration	Scale	Temp.	R^2	MAE	Time (s)	Errors
Gemini 3 Flash low thinking	8	0.0	0.4673	1.3098	669.5	154
Gemini 3 Flash low thinking	8	0.3	0.5019	1.2943	659.1	149
Gemini 3 Flash low thinking	8	0.7	0.5182	1.2620	666.4	155

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